



Semester 1 Examinations 2014 - 2015

Exam Code	1SD1, 1MDM1, 1IF1, 1MF1, 1MCR1, 1SD3
Exam	Higher Diploma in Software Design and Development Higher Diploma in Software Design and Development, Industry Stream Masters in Information Technology Masters in Software Design and Development Masters in Digital Media Masters Clinical Research
Module Code	CT511
Module	Database Development
External Examiner	Professor Liam Maguire
Internal Examiners	*Ms. Pat Byrne Professor G. Lyons Dr. Michael Madden
<u>Instructions</u>	Answer question 1 and any two others
Duration	2 hours
No. of Pages	3
Department	Information Technology
Requirements	None

1. (a) A small company selling gas stoves and heating systems requires a database to help manage its sales activities. The company knows that there is a wide variation in sales at different times of the year and the management wish to be able to analyse the seasonal income in order to schedule special sales and offers. The company employs four sales persons in their Galway showroom to demonstrate and sell products to the general public. Each of these works for a commission based on the cost of sales. While detailed information is not held about individuals who purchase, business is also done with building contractors who may wish to buy several stoves and fireplaces for homes they are building. These contractors are billed at the end of each month and have negotiated a trade discount for their purchases. The products sold come from five main suppliers, some of whom are overseas, and details need to be kept on shipping information so appropriate tax returns are made.

Draw up a set of normalised tables to meet their needs, indicating all primary and foreign keys and table relationships. Document any assumptions you make in your design.

[20]

- (b) The government body responsible for managing national museums maintains a database of the permanent and temporary exhibitions they offer. The following tables are part of this database:

MUSEUM	Mus_code	Mus_name	Mus_Locn	Mus_Co	
EXHIBIT	Exh_no	Exh_name	Mus_code	Exh_start	Exh_end
TICKETS	Tick-no	Tick-price	Exh-no	Tick-type	
EMPLOYEE	Emp_no	Emp-name	Emp_dob	Emp_Co	Emp_phone
WORK	Emp_no	Exh_no	Work_date	Work_hrs	

NOTE:

The MUSEUM table contains details of the museum including its location (Mus_loc) and the county in which it is situated (Mus_co)

The WORK table is a record of the hours worked by each employee at any exhibition.

The EMPLOYEE table has basic details for each employee, including their county of residence (Emp_co).

The TICKETS table has an entry for each ticket sold, including the type of ticket (C for child, A for adult)

Create SQL queries to satisfy the following information requests:

- i. List the total hours worked by each employee on each exhibition.
- ii. List the name and phone number of all employees who worked on 17th May 2014 in the “Irish Gold” exhibition.
- iii. Show the number of people who have visited each exhibition (tickets sold).
- iv. List all employees who have not worked on any exhibition.
- v. List all employees who have worked at an exhibition in the county in which they live.

[20]

2. (a) You have been asked to create a database for a small company of solicitors so that it might keep track of charging their clients (based on the number of hours consultation or in court). Describe how you might go about gathering all the information you require to carry out your database design. [10]
- (b) Explain the following terms when used with databases:
tuple ; metadata ; data mining ; denormalisation ; deadlock [20]
3. (a) Explain how you might decide the selection of primary and foreign keys in designing a database. Use a small example to illustrate your answer. [20]
- (b) Explain the function and use of a data dictionary. [10]
4. (a) Explain how a data warehouse differs from a relational database. How might you go about setting one up? [20]
- (b) In a distributed database, choices must be made about where the data is physically held on disk. Explain some of the techniques used to apportion data. [10]